LearnIQ

BE Students — Multiple Choice Question Paper

Purpose: Website testing and assessment practice

Total Questions: 40 **Maximum Marks:** 40 **Suggested Time:** 45 minutes

Instructions: Select one correct option for each question. Each question carries 1 mark. There is no negative marking.

Programming Languages (Java, Python, C++)

1. In Java, which statement about method overloading is correct?

A. It requires methods to have different return types only

B. It allows methods with the same name but different parameter lists

C. It can occur only between a parent and child class

D. It is resolved only at runtime

2. What is the output of Python expression `type(3 / 2).__name__` in Python 3?

A. int

B. float

C. decimal

D. rational

3. In C++, which feature allows a derived class to provide a specialized implementation of a base-class method?

A. Function overloading

B. Operator precedence

C. Function overriding

D. Namespace aliasing

4. Which Java keyword prevents a class from being inherited?

A. static

B. final

C. private

D. abstract

5. What is the average-case time complexity of searching for a key in a Python dictionary?

A. O(1)

B. O(log n)

C. O(n)

D. O(n log n)

6. In C++, what is the primary purpose of a virtual destructor in a polymorphic base class?

A. To prevent object creation

B. To make the class abstract

C. To ensure the derived destructor runs when deleting through a base pointer

D. To overload the delete operator

7. Which Python construct is commonly used to handle exceptions?

A. try-except

B. for-in

C. with-if

D. def-catch

8. Which Java collection does not allow duplicate elements?

A. ArrayList

B. LinkedList

C. HashSet

D. Vector

9. In C++, which cast is generally intended for safe conversions within an inheritance hierarchy using runtime type information?

A. static_cast

B. reinterpret_cast

C. const_cast

D. dynamic_cast

10. What does Python's `__init__` method typically do?

A. Destroys an object

B. Initializes a newly created object

C. Imports a module

D. Converts a class into a function

Operating Systems

11. Which scheduling algorithm can cause starvation of low-priority processes?

A. First-Come, First-Served

B. Round Robin

C. Priority Scheduling

D. FIFO page replacement

12. A process is in the blocked state when it is typically:

A. Waiting for an event such as I/O completion

B. Currently executing on the CPU

C. Finished execution

D. Newly admitted but not yet scheduled

13. Which condition is necessary for deadlock to occur?

A. Mutual exclusion

B. Circular wait

C. Hold and wait

D. All of the above

14. What is the main purpose of a Translation Lookaside Buffer (TLB)?

A. Store files permanently

B. Cache recent virtual-to-physical address translations

C. Schedule disk requests

D. Detect CPU overheating

15. Which page-replacement algorithm can exhibit Belady's anomaly?

A. LRU

B. Optimal

C. FIFO

D. None of these

Database Management Systems (DBMS)

16. Which ACID property ensures that a transaction is completed entirely or not at all?

A. Consistency

B. Isolation

C. Atomicity

D. Durability

17. A relation is in Third Normal Form (3NF) when, among other requirements, it has no:

A. Candidate keys

B. Transitive dependency of non-key attributes on a key

C. Foreign keys

D. Composite attributes in every row

18. Which SQL clause filters groups after aggregate functions are computed?

A. WHERE

B. ORDER BY

C. HAVING

D. DISTINCT

19. What does a foreign key primarily enforce?

A. Uniqueness in every column

B. Referential integrity between tables

C. Automatic indexing of all columns

D. Encryption of a table

20. Which index structure is commonly used by relational databases to support efficient range queries?

A. B-tree or B+ tree

B. Stack

C. Queue

D. Hash table only

Logical Reasoning and Aptitude

21. A machine completes a task in 12 hours. Assuming a constant rate, what fraction of the task does it complete in 3 hours?

A. 1/2

B. 1/3

C. 1/4

D. 1/6

22. Find the next number in the sequence: 3, 8, 15, 24, 35, __.

A. 42

B. 46

C. 48

D. 50

23. All engineers in a team know programming. Some programmers know AI. Which conclusion must be true?

A. All engineers know AI

B. Some engineers know AI

C. All programmers are engineers

D. No conclusion about engineers knowing AI necessarily follows

24. A bag contains 4 red, 5 blue, and 3 green balls. What is the probability of randomly selecting a blue ball?

A. 1/3

B. 5/12

C. 5/7

D. 3/12

25. If 5 workers finish a job in 18 days at the same rate, how many days would 6 workers take?

A. 12

B. 15

C. 18

D. 21

Object-Oriented Programming (OOP)

26. Which OOP principle bundles data and the methods operating on it while controlling access to internal state?

A. Encapsulation

B. Inheritance

C. Polymorphism

D. Compilation

27. When a subclass can be used wherever its superclass is expected without breaking correctness, this reflects:

A. Cohesion

B. Liskov Substitution Principle

C. Method hiding

D. Data duplication

28. Which relationship is best represented by composition?

A. A university has departments

B. A square is a rectangle

C. A dog is an animal

D. A class implements an interface

29. Which statement best distinguishes abstraction from encapsulation?

A. Abstraction exposes implementation details; encapsulation removes classes

B. Abstraction focuses on essential behavior; encapsulation bundles state and controls access

C. They are identical concepts

D. Encapsulation applies only to procedural programming

30. In runtime polymorphism in Java, which mechanism commonly selects the overridden instance method implementation?

A. Static binding at compile time

B. Dynamic method dispatch

C. Preprocessor expansion

D. Constructor chaining

Data Structures and Algorithms (Theory)

31. Which data structure is most appropriate for implementing a function-call stack?

A. Queue

B. Stack

C. Graph

D. Hash table

32. What is the worst-case time complexity of binary search on a sorted array?

A. O(1)

B. O(log n)

C. O(n)

D. O(n log n)

33. Which traversal of a binary search tree visits keys in non-decreasing order?

A. Preorder

B. Postorder

C. Inorder

D. Level order

34. Which collision-resolution technique stores colliding hash-table entries in a list or bucket at each index?

A. Separate chaining

B. Binary search

C. Merge sort

D. Path compression

35. Which algorithm is commonly used to find shortest paths from a single source in a graph with non-negative edge weights?

A. Kruskal's algorithm

B. Dijkstra's algorithm

C. Prim's algorithm

D. DFS only

36. What is the worst-case time complexity of merge sort?

A. O(n)

B. O(log n)

C. O(n log n)

D. $O(n^2)$

37. Which data structure is typically used by breadth-first search (BFS) to manage the frontier?

A. Stack

B. Queue

C. Heap only

D. Disjoint-set forest

38. In a max-heap, which property must hold for every parent node and its children?

A. Parent is less than both children

B. Parent is greater than or equal to its children

C. The tree is always a binary search tree

D. All leaves have the same value

39. What is the main advantage of an adjacency list over an adjacency matrix for a sparse graph?

A. It always gives O(1) edge lookup

B. It uses less space when the graph has relatively few edges

C. It sorts vertices automatically

D. It eliminates graph traversal

40. Which statement about a stable sorting algorithm is correct?

A. It always runs in O(n)

B. It uses no extra memory

C. It preserves the relative order of equal-key elements

D. It can sort only integers

Answer Key

Correct options for all 40 questions

Use this section for checking responses after completing the test.

Question	Answer	Question	Answer
1	B	21	C
2	B	22	C
3	C	23	D
4	B	24	B
5	A	25	B
6	C	26	A
7	A	27	B
8	C	28	A
9	D	29	B
10	B	30	B
11	C	31	B
12	A	32	B
13	D	33	C
14	B	34	A
15	C	35	B
16	C	36	C
17	B	37	B
18	C	38	B
19	B	39	B
20	A	40	C

Section distribution: Programming Languages (10), Operating Systems (5), DBMS (5), Logical Reasoning & Aptitude (5), OOP (5), DSA Theory (10).